

Memorial University of Newfoundland
CMSC 6950: Computer Based Research Tools and Applications
Tentative Course Schedule – Fall 2025

Class	Topic	Notes
Sept. 9	Introduction to the Unix shell	HW1 out
Sept. 11	Advanced command line and shell scripting	
Sept. 16	Introduction to Python and notebooks	
Sept. 18	Python libraries and modules; scripts	
Sept. 23	Introduction to Git	
Sept. 25	Collaborating with Git and Github	HW2 out
Sept. 30	No class; National Day for Truth and Reconciliation	
Oct. 2	Reproducibility	HW1 due
Oct. 7	pytest	HW3 out
Oct. 9	Continuous Integration / Continuous Delivery	
Oct. 14	No class; Thanksgiving	
Oct. 16	Virtual environments; Python dataframes	HW2 due
Oct. 21	Python lists; loops; conditionals	Project proposals due
Oct. 23	Python plotting; working with data sets	HW4 out
Oct. 28	Advanced plotting	HW3 due
Oct. 30	Data visualization	HW5 out
Nov. 4	Creating Python functions; variable scope	
Nov. 6	Programming style conventions and linters	
Nov. 11	No class; Remembrance Day	
Nov. 12	Debugging; scripts and the command line	Tuesday schedule
Nov. 13	Numpy package	HW6 out
Nov. 18	Scipy, part 1: integrate and interpolate	HW5 due
Nov. 20	Scipy, part 2: linalg and optimize	HW7 out
Nov. 25	Profiling code	HW6 due
Nov. 27	Making code run faster: Numba and pybind/cython	
Dec. 2	Final project presentations	HW7 and projects due
Dec. 4	Final project presentations	

Notes:

- Changes to class schedule will be announced via Brightspace, including cancellations or transitions to online discussions.
- If you need to miss class for any reason, I can provide specific readings for any material missed.